

## MEKNES, Morocco

WMO: 601500

Lat: 33.879N

Lon: 5.515W

Elev: 576

StdP: 94.59

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 2.6        | 3.8 | -4.1                          | 2.9 | 11.7 | -2.2 | 3.4 | 11.1 | 9.6                   | 11.1 | 8.5 | 10.8 | 2.9                   | 160  | 0.414 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 14.8                      | 39.4            | 20.4 | 37.1 | 20.4 | 35.1 | 20.2 | 23.6                | 33.4 | 22.4 | 32.2 | 21.6 | 31.4 | 3.9                     | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.2                            | 16.0 | 29.3 | 19.2 | 15.0 | 27.2 | 18.2 | 14.1 | 25.5 | 73.3          | 33.5 | 68.8 | 32.3 | 65.4 | 31.1 | 30.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| 8.5               | 7.3  | 6.3 | DB                         | 0.0  | 42.5               | 1.7 | 1.7                                                | -1.2 | 43.7       | -2.2 | 44.7       | -3.1 | 45.6       | -4.4 | 46.8 | (4) |
|                   |      |     | WB                         | -1.3 | 25.2               | 1.9 | 2.5                                                | -2.7 | 27.0       | -3.8 | 28.5       | -4.8 | 29.9       | -6.2 | 31.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 18.0  | 10.6  | 11.4  | 13.8  | 15.7  | 19.3  | 23.3  | 25.9  | 26.3  | 22.9  | 19.9  | 14.7  | 11.6  | (6)  |
|      |                                                                   | DBStd       | 6.36  | 2.51  | 2.79  | 3.08  | 3.10  | 3.97  | 3.75  | 3.52  | 3.29  | 3.01  | 3.32  | 3.43  | 2.49  | (7)  |
|      |                                                                   | HDD10.0     | 57    | 22    | 17    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 12    | 12    | (8)  |
|      |                                                                   | HDD18.3     | 1056  | 240   | 195   | 143   | 92    | 35    | 2     | 0     | 2     | 22    | 118   | 208   | 208   | (9)  |
|      |                                                                   | CDD10.0     | 2971  | 41    | 55    | 123   | 171   | 287   | 398   | 493   | 505   | 387   | 306   | 143   | 62    | (10) |
|      |                                                                   | CDD18.3     | 927   | 0     | 0     | 4     | 12    | 63    | 149   | 235   | 247   | 139   | 69    | 9     | 0     | (11) |
|      |                                                                   | CDH23.3     | 10699 | 1     | 4     | 69    | 190   | 797   | 1767  | 2890  | 2968  | 1308  | 628   | 77    | 2     | (12) |
| (14) | Wind                                                              | CDH26.7     | 5413  | 0     | 1     | 11    | 48    | 361   | 907   | 1632  | 1658  | 572   | 214   | 10    | 0     | (13) |
|      |                                                                   | WSAvg       | 2.8   | 2.9   | 3.0   | 3.2   | 3.1   | 2.9   | 2.8   | 2.5   | 2.6   | 2.6   | 2.7   | 2.9   | 3.0   | (14) |
|      |                                                                   | PrecAvg     | 568   | 74    | 71    | 68    | 58    | 38    | 12    | 2     | 18    | 49    | 70    | 76    | 76    | (15) |
|      |                                                                   | PrecMax     | 1019  | 240   | 226   | 178   | 222   | 135   | 59    | 21    | 13    | 126   | 195   | 183   | 261   | (16) |
|      |                                                                   | PrecMin     | 347   | 0     | 0     | 1     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |
|      |                                                                   | PrecStd     | 166   | 60    | 51    | 48    | 49    | 32    | 14    | 4     | 3     | 24    | 44    | 49    | 58    | (18) |
|      |                                                                   | DB          | 22.2  | 23.2  | 28.1  | 30.8  | 37.1  | 40.2  | 42.1  | 41.9  | 38.1  | 33.2  | 28.1  | 22.9  | 22.9  | (19) |
| (20) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | MCWB        | 12.4  | 13.8  | 16.2  | 16.9  | 18.3  | 19.6  | 20.2  | 22.5  | 19.1  | 16.7  | 15.1  | 12.8  | 12.8  | (20) |
|      |                                                                   | DB          | 19.7  | 21.1  | 25.1  | 27.9  | 33.8  | 37.1  | 39.4  | 39.2  | 34.9  | 31.1  | 25.6  | 20.7  | 20.7  | (21) |
|      |                                                                   | MCWB        | 11.9  | 13.2  | 15.2  | 16.4  | 18.2  | 19.7  | 20.6  | 20.8  | 19.5  | 16.9  | 14.4  | 12.6  | 12.6  | (22) |
|      |                                                                   | DB          | 17.9  | 19.2  | 23.1  | 25.4  | 30.9  | 34.4  | 37.0  | 36.9  | 32.0  | 29.1  | 23.2  | 18.7  | 18.7  | (23) |
|      |                                                                   | MCWB        | 11.5  | 12.9  | 14.7  | 16.0  | 17.8  | 19.8  | 20.7  | 21.0  | 19.5  | 16.9  | 13.9  | 12.3  | 12.3  | (24) |
|      |                                                                   | DB          | 16.0  | 17.3  | 20.9  | 22.9  | 27.9  | 31.9  | 34.2  | 34.2  | 29.9  | 27.0  | 21.1  | 16.9  | 16.9  | (25) |
|      |                                                                   | MCWB        | 11.1  | 12.1  | 14.2  | 15.5  | 17.2  | 19.3  | 20.6  | 20.6  | 18.9  | 16.7  | 13.8  | 11.7  | 11.7  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | WB          | 14.7  | 15.9  | 18.2  | 19.7  | 21.3  | 23.1  | 25.5  | 26.3  | 22.7  | 20.2  | 17.8  | 15.7  | 15.7  | (27) |
|      |                                                                   | MCDB        | 19.0  | 21.0  | 24.8  | 26.5  | 31.9  | 33.5  | 35.7  | 37.8  | 31.1  | 27.0  | 21.8  | 19.2  | 19.2  | (28) |
|      |                                                                   | WB          | 13.4  | 14.5  | 16.7  | 18.0  | 19.8  | 21.9  | 23.6  | 23.7  | 21.4  | 19.1  | 16.5  | 14.4  | 14.4  | (29) |
|      |                                                                   | MCDB        | 17.4  | 19.3  | 22.7  | 25.0  | 28.6  | 32.6  | 33.6  | 33.5  | 30.0  | 25.8  | 21.6  | 17.6  | 17.6  | (30) |
|      |                                                                   | WB          | 12.5  | 13.5  | 15.6  | 16.8  | 18.6  | 20.8  | 22.3  | 22.5  | 20.5  | 18.4  | 15.5  | 13.4  | 13.4  | (31) |
|      |                                                                   | MCDB        | 16.1  | 17.9  | 21.1  | 23.2  | 27.6  | 30.8  | 32.4  | 32.0  | 28.4  | 24.9  | 20.6  | 16.7  | 16.7  | (32) |
|      |                                                                   | WB          | 11.5  | 12.6  | 14.5  | 15.7  | 17.7  | 19.9  | 21.3  | 21.6  | 19.8  | 17.6  | 14.5  | 12.5  | 12.5  | (33) |
| (34) | Mean Daily Temperature Range                                      | MCDB        | 14.9  | 16.5  | 19.7  | 21.7  | 26.4  | 29.5  | 31.5  | 31.2  | 27.4  | 23.9  | 19.4  | 15.9  | 15.9  | (34) |
|      |                                                                   | MDBR        | 9.7   | 10.5  | 11.3  | 11.8  | 12.9  | 14.4  | 15.1  | 14.8  | 12.7  | 11.5  | 10.1  | 9.8   | 9.8   | (35) |
|      |                                                                   | 5% DB       | 12.8  | 13.9  | 15.3  | 16.3  | 17.5  | 17.6  | 17.1  | 17.0  | 15.7  | 14.8  | 13.2  | 12.4  | 12.4  | (36) |
|      |                                                                   | MCWBR       | 7.7   | 8.4   | 8.3   | 8.0   | 7.0   | 6.4   | 6.6   | 6.6   | 6.1   | 6.0   | 6.3   | 7.2   | 7.2   | (37) |
|      |                                                                   | 5% WB       | 10.7  | 12.0  | 13.2  | 14.2  | 15.1  | 15.6  | 15.4  | 15.3  | 13.4  | 12.1  | 10.6  | 10.2  | 10.2  | (38) |
|      |                                                                   | MCWBR       | 7.2   | 7.8   | 7.9   | 7.9   | 7.0   | 6.7   | 7.2   | 7.1   | 6.0   | 5.8   | 5.7   | 6.7   | 6.7   | (39) |
|      |                                                                   | taub        | 0.297 | 0.331 | 0.381 | 0.389 | 0.419 | 0.453 | 0.497 | 0.478 | 0.423 | 0.384 | 0.335 | 0.307 | 0.307 | (40) |
| (41) | Clear-Sky Solar Irradiance                                        | taud        | 2.537 | 2.430 | 2.274 | 2.274 | 2.210 | 2.105 | 1.966 | 2.033 | 2.220 | 2.339 | 2.476 | 2.538 | 2.538 | (41) |
|      |                                                                   | Ebn at Noon | 923   | 918   | 891   | 898   | 870   | 836   | 797   | 806   | 837   | 845   | 866   | 889   | 889   | (42) |
|      |                                                                   | Edn at Noon | 81    | 101   | 128   | 134   | 144   | 160   | 183   | 168   | 133   | 109   | 86    | 77    | 77    | (43) |
|      |                                                                   | RadAvg      | 3.00  | 3.84  | 4.99  | 5.95  | 6.75  | 7.69  | 7.72  | 6.96  | 5.67  | 4.26  | 3.17  | 2.77  | 2.77  | (44) |
| (45) | All-Sky Solar Radiation                                           | RadStd      | 0.33  | 0.40  | 0.40  | 0.41  | 0.52  | 0.34  | 0.16  | 0.24  | 0.29  | 0.31  | 0.28  | 0.26  | 0.26  | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only           | N/A   | N/A     | N/A    | +1.30   | -1.03 | -1.12 | N/A         | N/A     | +111    | +160    | (46) |  |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | +1.05   | N/A   | -0.48 | N/A         | N/A     | N/A     | +81     | (47) |  |

Nomenclature: See separate page